Sindy's [VP240] Final

32 Electromagnetic Waves

Visible lights: 380(red) – 750(violet) nm
Electromagnetic waves are transverse: direction of propagation is same as $\vec{E} \times \vec{B}$.

$$E = cB,\ B = \epsilon_0\mu_0 cE,\ c = 1/\sqrt{\epsilon_0\mu_0}$$

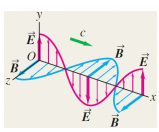
Energy flow rate: (power per unit area)

$$\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B},\ I = S_{av} = \frac{E_m^2}{2\mu_0 c}$$

Momentum flow rate:

$$\frac{1}{A}\frac{dp}{dt} = \frac{S}{c} = \frac{EB}{\mu_0 c}$$

Standing Waves: Perfect reflecting surface
Nodal(de) planes for E: $kx = 0, \pi, 2\pi, ...$
Nodal(de) planes for B: $kx = \pi/2, 3\pi/2, ...$



33 The Nature of Light

Personalities of Light:
1. Particle property: Energy carried is packaged in discrete bundles of photons.
2. Wave property: Propagation described by a wave model.
3. Emitted by accelerated electric charges.
Wave front: surface of constant phase.
Ray: line along direction of propagation.

When light is transmitted from one material to another, (n – index of refraction)

$$n = c/v \geq 1, \quad \lambda = \lambda_0/n$$

Reflection and Refraction

Specular reflection vs. Diffuse reflection (scattered reflection from a rough surface)

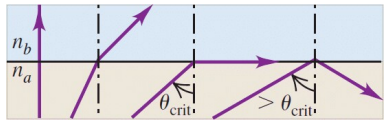
The laws of reflection and refraction:
1. Incident, reflected, refracted rays, normal to surface lie in plane of incidence.
2. law of reflection: $\theta_r = \theta_a$
3. law of refraction: $n_a \sin\theta_a = n_b \sin\theta_b$ (Snell's law)

Total Internal Reflection

When a ray travels in a material a to another b, ($n_a > n_b$), total internal reflection occurs when $\theta > \theta_{crit}$

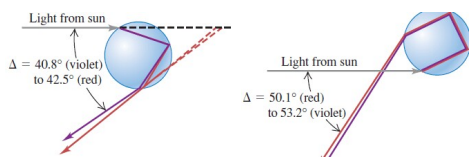
$$\sin\theta_{crit} = \frac{n_b}{n_a}$$

Application: Porro Prism; light pipe; optical fibers (endoscope,communication (fast, distinct)); diamond



Dispersion

Violet deviated most; red deviated least; others in intermediate positions. The light dispersed into a spectrum.
Rainbows: Dispersion, refraction, reflection.
Light from behind reflected from back surface of drop, refracted upon exiting. A light ray entering middle reflected straight back. Others exit within angle Δ of middle ray, with rays piling up. A disk of light of angular radius centered on down-sun point. Disk bright around its rim; sky dark outside (no light reaches from angle larger than Δ)
Second rainbow: Fainter. Reversed Sequence.

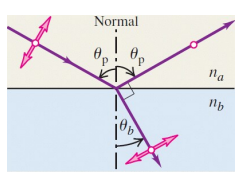


Polarization

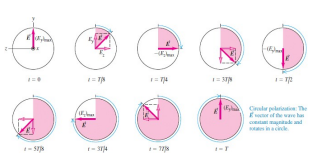
Direction of polarization is same as $\vec{E}$.
Malus's law $I = I_{max}\cos^2\phi$
Polarizing by reflection: When light strikes interface, Brewster's law states reflected light completely polarized perpendicular to plane of incidence, if $\theta > \theta_p$.

$$\tan\theta_p = n_b/n_a$$

Application: sunglasses (axis vertical)



Circular/Elliptical Polarization.

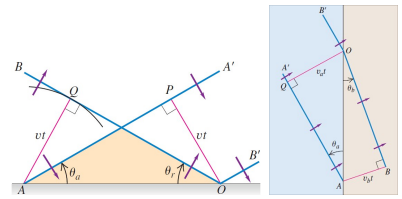


Scattering of Light

Why sky blue? Scattered light contains more blue light than red.
Why sunset red? Light travel long in atmosphere, blue light moved by scattering.
Why clouds/milk white? High concentration drop/fat globules, all lights scattered out of cloud.

Huygens's Principle

Points of a wave front considered source of 2nd wavelets that spread out in all directions.



Why mirages? When surface of sand heated, less dense ($n \downarrow$, $v \uparrow$) layer of air forms. Huygens wavelets have $\uparrow$ radii, wave font tilt, ray that were headed toward surface with large θ_a be bent up. Light farther from ground is bent less.

34 Geometric Optics

Object: point, extended – radiate light
Image: virtual, real; erect, inverted; (reversed)

Focal point F; Vertex V;
Focal length f (d_{FV})
Concave (ao); Convex (tu); Optic axis;
Center C/ Radius R of curvature
Lateral magnification $m = y'/y$ (im/ob)

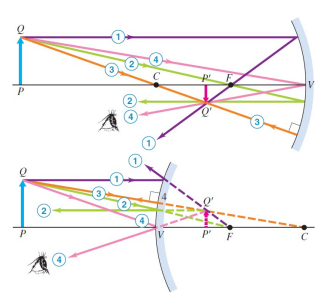
Sign rules:
1. object distance s: + if on incoming side(real).
2. image distance s': + if on outgoing side(real).
3. R: + if C is on outgoing side.

Graphical method – principle rays
1. ray parallel (reflected, through F).
2. ray through F (reflected parallel).
(3. ray along radius, through C.)
4. ray to V (reflected forming eq θ).

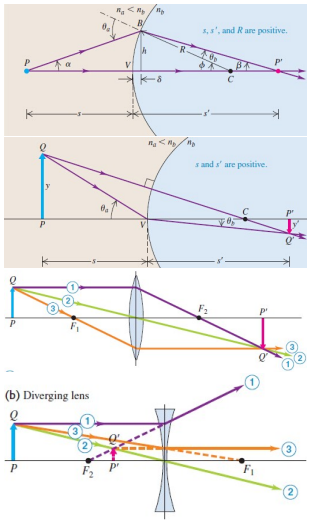
Image formed by one surface/optical device can serve as object for a second one.

Thin lenses[1] $1/s + 1/s' = 1/f$
Lensmakers' equation

$$1/f = (n-1)(1/R_1 - 1/R_2)$$



[1] paraxial rays, paraxial approximation, spherical aberration



Application

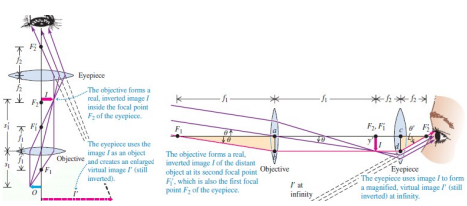
Camera capability of light-gathering:

$$f - number = f/D_{Aperture}$$

Eyes: cornea, retina; near/farsighted
Magnifier: angular magnification

$$M = \frac{\theta'}{\theta_{25\ \mathrm{cm}}} = \frac{25\ \mathrm{cm}}{f}$$

Microscopes and Telescopes:
Primary lens (object) form real image, second lens (eyepiece) used as a magnifier to make enlarged, virtual image.



35 Interference

Monochromatic light: single frequency
Coherence: a unchanging phase relation

Two-source Interference of Light

Young two-slit experiment, source in phase
Constructive interference, (antinodal)

$$r_2 - r_1 = d\sin\theta = m\lambda\ (m \in \mathbb{Z})$$

Destructive interference, (nodal curves)

$$r_2 - r_1 = d\sin\theta = (m + 1/2)\lambda\ (m \in \mathbb{Z})$$

If small angles $y = R(r_1 - r_2)/d = R\sin\theta$. The close together the slits are, the more the pattern spread out.

Intensity in Interference Pattern

$$\text{Amplitude } E_p = 2E|\cos(\phi/2)|$$
$$\text{Intensity } I = I_0\cos^2(\phi/2)$$
$$\text{Phase diff vs. path diff (k – wave number)}$$
$$\phi = (2\pi/\lambda)(r_2 - r_1) = k(r_2 - r_1)$$

Interference in Thin Films

Why colored rings or fringes?
Light shining on upper surface of thin film with thickness t partly reflected at upper surface. Light transmitted through upper surface partly reflected at lower surface. Two reflected waves come together at point P on retina of eyes. Depending on phase relation, they interfere constructively or destructively. Different colors, different wavelengths, so interference is constructive for some and destructive for others.

From Maxwell's equations, (normal incidence) $E_r = E_i(n_a - n_b)/(n_a + n_b) \Rightarrow$
1. $n_a > n_b$ no phase shift
2. $n_a = n_b$ no reflection
3. $n_a < n_b$ half-cycle phase shift

If neither/ both have half-cycle shift

$$2t = m\lambda\ (m \in \mathbb{N})\ \text{con}$$
$$2t = (m + 1/2)\lambda\ (m \in \mathbb{N})\ \text{de}$$

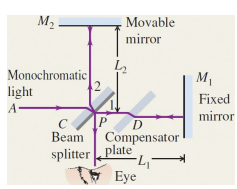
If one of them has half-cycle shift

$$2t = (m + 1/2)\lambda\ (m \in \mathbb{N})\ \text{co}$$
$$2t = m\lambda\ (m \in \mathbb{N})\ \text{de}$$

Thin: to ensure coherent wave
Air wedge; Newton's Ring

Nonreflective coating: $t = \frac{1}{4}\lambda$, $n_{air} < n < n_{glass}$ (e.g. camera lens)
Reflective coating: $t = \frac{1}{4}\lambda$, $n > n_{glass}$ (e.g.infrared "heat reflectors")

Michelson Interferometer: measures λ: $y = m\frac{\lambda}{2}$



36 Diffraction

Fresnel (near) vs. Fraunhofer (far) diffraction

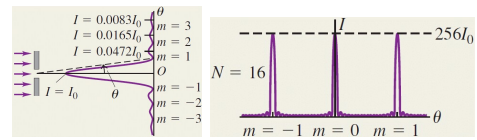
Single-slit Diffraction

Destructive interference

$$\sin\theta = m\lambda/a\ (m = \pm1, \pm2, ...)$$

Intensity as a function of θ

$$I = I_0\left\{\frac{\sin[\pi a(\sin\theta)/\lambda]}{\pi a(\sin\theta)/\lambda}\right\}^2$$



Diffraction Grating

Consists of many thin parallel slits, spaced distance d apart. Condition for maximum intensity same as for two-source pattern, but maxima sharp and narrow.

$$d\sin\theta = m\lambda\ (m = 0, \pm1, \pm2, ...)$$

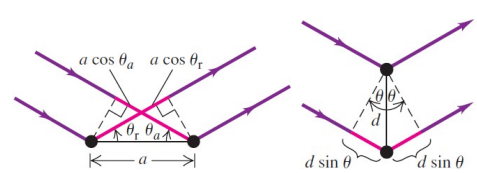
Why DVD colorful? Reflection grating, array of ridges/ grooves on reflective screen. Reflected light, maximum intensity at angles where, adjacent $2k\pi$. Information coded on DVD by varying length of pits.

X-ray diffraction

Crystal serves as 3-D diffraction grating for x-ray (λ same order of magnitude as spacing between atoms). For a set of crystal planes spaced d apart, con interference occurs when angles of incidence and scattering are equal and when Bragg condition ($2d\sin\theta = m\lambda\ (m \in \mathbb{N}^*)$) is satisfied.

Circular Apertures & Resolving Power

Diffraction pattern form a circular aperture of diameter D consists of a central bright spot (Airy disk) and a series of concentric dark and bright rings. For 1st dark ring $\sin\theta_1 = 1.22\lambda/D$ (angular size of Airy disk). Diffraction sets limit on resolution (image sharpness) of optical instruments. According to Rayleigh's criterion, two points are just resolved when their angular separation $\theta = \theta_1$



Plane Mirror	Spherical Mirror	Plane Refracting Surface	Spherical Refracting Surface
$1/s + 1/s' = 0$	$1/s + 1/s' = 2/R = 1/f$	$n_a/s + n_b/s' = 0$	$n_a/s + n_b/s' = (n_b - n_a)/R$
$m = -s'/s = 1$	$m = -s'/s$	$m = -(n_a s')/(n_b s) = 1$	$m = -(n_a s')/(n_b s)$